

Fig. 6: Actual-size PCB layout of the transmitter circuit

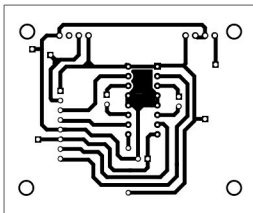


Fig. 8: Actual-size PCB layout of the receiver circuit

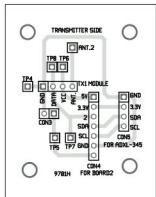


Fig. 7: Component layout of the transmitter circuit

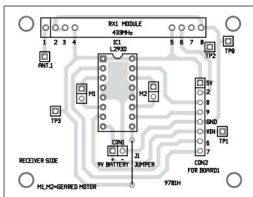


Fig. 9: Component layout of the receiver circuit

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

void writeTo(int device, byte address, byte val). This function is used to write bytes of data from a particular byte address to the slave device.

void readFrom(int device, byte address, int num, byte buff[]). This function is used to read bytes of data of particular buff sizes from a specific buffer address.

void setup(). This function is used to initialise the baud rate for serial communication and other pins for digital input/output.

void loop(). This function is used to perform the required task of comparing data with the reference set by the user for X, Y and Z axes and perform the task of transmitting data

for forward, backward, left and right movement of the robot.

Construction and testing

An actual-size, single-side PCB layout of the transmitter circuit for the hand-gesture-controlled robot is shown in Fig. 6 and its component layout in Fig. 7. An actual-size, single-side PCB layout of the receiver circuit is shown in Fig. 8 and its component layout in Fig. 9. Connect Board1 to CON2 in the PCB layout (Fig. 9), Board2 to CON4 and ADXL345 to CON5 in the PCB layout (Fig. 7). Pin numbers shown in CON2 and CON4 refer to the digital pins of Arduino boards.

Fit the transmitter circuit in a glove so that it can be used repeatedly. Place ADXL345 below the middle finger such that the pins' side of ADXL345 is facing towards you as shown in Fig. 10. We use the middle finger for attaching ADXL345 because it will get the best average



Fig. 10: Attaching the accelerometer module between fingers

PARTS LIST

Semiconductors:

- LCD1 - 16×2 LCD
- IC1 - L293D motor driver
- BOARD1, BOARD2 - Arduino UNO board
- BOARD2 - 433MHz RF transmitter module
- TX1 - 433MHz RF receiver module
- RX1 - 433MHz RF receiver module
- ADXL345 - Accelerometer module

Miscellaneous:

- M1, M2 - 6V, DC geared motors
- CON1, CON3 - 2-pin connector
- CON2 - 8-pin connector
- CON4 - 7-pin connector
- CON5 - 4-pin connector
- ANT.1, ANT.2 - 30cm-long single-strand wire antenna
- USB A-B cable
- 9V battery (2)

motion of the movement of the hand.

The rest of the circuit of Arduino and RF transmitter can be fitted near your wrist or on a separate PCB so that it does not disturb hand movements.

On the receiver side of the robot, Arduino UNO board should be connected to L293D IC and the remaining circuit through proper connection of wires. We have used standard 9V batteries to power transmitter and receiver circuits during testing. Use of separate voltage sources for Arduino UNO board and L293D motor driver IC is recommended for better performance. A 30cm-long wire can be attached to antenna pins of both transmitter and receiver modules for better communication. ●



Prashant Kumar is a B.Tech from IIT, Jodhpur. His interests include working with robotics, circuit design and Arduino boards